

Thermochemical Upgrading of Calcium-Bentonite for Drilling Fluid Applications

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ABSTRACT—The rheological properties of bentonite suspensions depend on the chemical composition and the contained dominant element, such as calcium (Ca), potassium (K), and sodium (Na). Na-bentonite type is the one used in drilling fluids, because it has good dispersion stability, high swelling capacity, and outstanding rheological properties. Ca-bentonite generally has bad rheological performance, however it can be activated by sodium to be used in drilling fluids, since there are huge unutilized Ca-bentonite resources. Many previous attempts of activation of Ca-bentonite were not feasible, upgrading required addition of many extra additives or sometimes mixed with commercial Na-bentonite to improve its properties. In this paper, a process of integrated beneficiation method is designed to efficiently remove the non-clay impurities and produce pure Ca-bentonite. An upgraded Ca-bentonite was produced using a combined thermochemical treatment in a wet process by adding 4wt.% of soda ash (Na_2CO_3) while heating and stirring. The new thermal treatment optimized and described in this study greatly improved the sodium activation and ions exchange process and improved bentonite properties. The thermochemically-upgraded Ca-bentonite outperformed the rheological properties of the commercial bentonite. And when tested in a typical drilling fluid formulation at high temperature, the investigations showed an identical behavior of the commercial drilling grade bentonite. Moreover, the results obtained showed that the thermochemically upgraded Ca-bentonite has higher yield point/plastic viscosity (YP/PV) ratio than commercial Na-bentonite when mixed with the drilling fluid additives. Higher YP/PV ratio is expected to enhance the hole cleaning and prevent most of the drilling problems.

Keywords: bentonite, drilling fluid, bentonite upgrading, soda ash, rheological properties.

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1.0 INTRODUCTION

Bentonite is widely used in drilling of oil and gas wells, due to its unique properties, swelling capacity, and ability to develop stable suspensions with exceptional rheological properties. It was firstly introduced as a drilling fluid additive in the early twentieth century, the first type used was the Gumbo clay [1]. Many different types of clay additives were used to formulate water based drilling fluids [2, 3]. A Good quality bentonite adsorbs water and swells, therefor; when used in drilling fluids, it gives the required rheological performance and ensures acceptable filtration loss [4, 5, 6, 7].

For successful drilling operations, well integrity and stability must be maintained. Also, a sufficient hydrostatic pressure should prevent formation influx from entering the wellbore, while filtration control reduces water filtrates from the drilling fluid to the adjacent formation. The circulation of the drilling fluid will lift the cutting from the bottom of the well to the surface. The most important hole cleaning parameter related to drilling fluid is the suspension capacity, which is the ability of drilling fluid to lift the cuttings and prevent settling while drilling or backing the well during static conditions or no operation time. This suspension capacity is directly related to the rheological properties. Viscosity and gel strength of the drilling fluid should be enough for hole cleaning by lifting the cutting from the bottom of the well up to the surface, and cutting suspension while drilling, tripping, or new connection [8].

1.1 Drilling fluids requirements and types of bentonites

Bentonite in nature can be divided into three groups: Naturally Na-bentonites, Naturally Ca-bentonites, and Na-activated Ca-bentonites [9]. The Na-bentonite has more ability to absorb water, significantly increase viscosity, and form thixotropic suspensions with good gel strength. In the other hand, Ca-bentonite has very low swelling capacity, and forms unstable suspension with high settling rates [10]. In drilling fluids, bentonite with good suspension property is required to make stable drilling fluids formulas, which can withstand various drilling operations and conditions. Stability of the drilling fluid formula will maintain the original properties of drilling fluid inside the well, under high formation temperature, and different shear rates applied by the drilling string. Moreover, drilling fluids face more challenges in deep drilling and unconventional resources, due to associated high pressures and high temperatures, HP/HT. Drilling fluids formula must be designed to provide good temperature dependent rheology profile, high temperature raises concerns about stability of the structure of the drilling fluids formula under harsh temperature environment. Therefore, rheological properties under this compression and expansion need to be maintained [11].

Drilling fluids should be designed to reduce the drilling fluid invasion that may cause formation damage.

Such damage in the near-wellbore may require special treatments to restore the well productivity. Jingbin et al. [12] introduced the radial jet drilling (RJD) technology to remove the damage in oil and gas wells. One major problem encountered in this method is the drag forces and hose resistance for bending.

Sodium bentonite is added to the drilling fluid to provide good rheological properties. Other additives such as polymers, starch, etc. are used to reduce the fluid loss of the drilling fluid. Recently, date seeds were investigated as a drilling fluid loss reducer. Adewole and Najimu [13] found out that the date seeds reduced the fluid loss of water-based drilling fluids.

The drilling fluid design depends on the well conditions. High temperature wells need different additives compared to low temperature well. Zhao et al. [14] investigated the use of polyalcohol and polyetheramine to reduce the fluid loss and inhibit shale swelling during drilling low temperature wells in the permafrost. The rheological properties of drilling fluid are highly affected by the reservoir mineralogical composition. Drilling fluids used to drill sandstone reservoirs will be contaminated with fine sand particles. Fine sand particles affected both the rheological and filtration properties of the drilling fluid [15, 16].

1.2 Quality check for drilling grade bentonite

Bentonite in nature is likely to contain many non-clay impurities, such as sand, quartz, feldspar, or iron. Purification or beneficiation of bentonite is a process of removing these contaminations and extracting the pure montmorillonite. This process of impurities removal can be obtained through chemical or mechanical methods, through separation techniques depend on the factors that affecting suspension property of bentonite, such as pH and particles size. Since bentonite is recognized as originally derived from volcanic ash, its composition should contain specific elements [17], these identifying elements are known as the clay minerals, which are used as a distinguishing parameter to characterize montmorillonite clays. According to API specification, the drilling fluids formula should satisfy the requirements of drilling operations condition in all aspects, such as filtration properties, mud cake thickness, mud rheological properties, and thermal stability. These properties can be evaluated through laboratory tests designed specifically to simulate the downhole condition of the drilling operation, based on API specifications [18]. Quality of bentonite for drilling fluid application is tested based on API specifications listed in API13-A for drilling fluid materials, where drilling grade bentonite is defined as a clay mineral from smectite group, that is chemically synthesized to meet OCMA specifications shown in the following **Table 1**.

After purification, bentonite, without further treatment; cannot be used directly as drilling fluid additives, even with high content of montmorillonite, bentonite may not be adequate to provide the acceptable viscosity values and the required level of filtration control [19]. Therefore, in most cases, bentonite properties are improved through various treatments methods, such as ionic liquids, polymer and inorganic

additives, to enhance its performance [20, 21, 22]. Sodium content greatly affects dispersion characteristics of bentonite. For drilling fluid, the swelling characteristics of Ca-bentonite can be enhanced significantly by soda ash treatment, where calcium ions are replaced with sodium ions to upgraded Ca-bentonite to Na-bentonite, and increasing sodium to calcium ratio [4]. Basically, even good quality bentonite, with high montmorillonite content ($> 90\%$), will still needs additional processing to fulfil the requirements of drilling operation. Many studies reported that, few amount of soda ash (Na_2CO_3) of approximately 2-4% can significantly increase the quality of bentonite. [7]. Aghamelu and Okogbue [20] reported that, with 4 wt. % Na_2CO_3 , most of the swellability and rheological parameters of local raw Ca-bentonite have improved. The specific gravity of the clays increased significantly from 2.42 to 2.90 with the addition of 4% barite as weighing agents. However, the improvement, especially in the swellability quality, is considerably very low when compared with swellability data on naturally active Na-bentonites [20].

1.3 Evaluation of local Saudi bentonite sources

Very few studies were reported on Saudi Ca-bentonites, mineral and chemical composition of different sources of Ca-bentonite was evaluated for possible use in drilling fluids [21]. However, the analysis showed that for drilling application, treatment with sodium carbonate, up to 5 wt.% did not upgrade bentonite properties to meet drilling fluids requirements. The Ca-bentonite must be improved by adding some other materials such as xanthan gum or carboxymethyl cellulose. Other studies succeeded in achieving 60% to 80% of the required properties by adding small quantities of commercial drilling grade bentonite with 3 to 6 wt.% sodium carbonate [21].

The ultimate objective of this study is to perform thermochemical upgrading of abundant available Ca-Bentonite for drilling fluid formulation. The local Ca-bentonite used in this study was fully characterized and tested to define the gap in performance when compared with commercial drilling grade Na-bentonite. A pure Ca-bentonite was extracted through simple and efficient purification process to remove all non-clay impurities, and the chemical and physical properties were identified. The sodium activation was carried out by new proposed thermo-chemical upgrading process [22], which proved to add a great impact on the activation process. The experimental work included measuring of particle size distribution, suspension stability, filtration, and the rheological properties at low and HP/HT in water-based bentonite mud and in selected real drilling fluids formulas.

2.0 MATERIALS AND EXPERIMENTAL METHODS

2.1 Materials

The Ca-bentonite was collected from local outcrop located in Jeddah, South-Western region of Saudi Arabia. The raw bentonite (As-received) was dark greenish brown color, the structure of the rock has clear uniform layered beds. Some gypsum, dirt, oxidization and weathering were observed due to erosion effect. Little black spots of manganese oxide also were clearly observed in many samples. The pure clay was extracted through an integrated purification process to remove impurities and non-clay elements, the term 'raw Ca-bentonite' will be used for the purified Ca-bentonite, and the term chemically treated Ca-bentonite will be used for activated Ca-bentonite through soda ash treatment only, while the term "upgraded Ca-bentonite" refers to the thermochemically treated raw Ca-bentonite. A commercial Na-bentonite was obtained from Sigma Aldrich with average particles size 2 to 4 μm and specific gravity 2.6. The composition and properties of the commercial bentonite was analyzed and used as a reference of comparison for the upgraded Ca-bentonite.

2.2 Methodology

Most of the reported bentonite beneficiation or purification methods involves one or some of these techniques; grinding, sieving (wet or dry), dispersion into deionized water, sedimentation, calcination, or uses some chemicals for extractions or enhancing clay suspension properties such as Sodium hexa-meta phosphate or pH manipulation materials [23]. In this study, for obtaining a pure raw Ca-bentonite, many purification techniques were tried out, to remove impurities and non-clay content from the samples. The best technique was selected in term of effectiveness, recovery percentage, cost, and time. The obtained raw Ca-bentonite then was tested according to API recommended practice 13-A, to define the gap in properties between Ca-bentonite and commercial Na-bentonite. They key parameters were; particles size measured using laser scattering particle size analyzer (Analysette 22 NanoTec, 0.02-2000 μm , Fritsch, France), Na/Ca ratio calculated from EDX analysis, suspension stability, rheological properties measured using discovery hybrid rheometers (DR-3 TA instrument, UK), and API filtration. For Sodium activation of Ca-bentonite, at the beginning, an ordinary chemical treatment with soda ash was conducted, and when improvement was not sufficient for drilling fluids, a novel thermochemically upgrading technique was developed [22].

2.3 Bentonite purification and characterization

Physical and chemical characteristic of the raw Ca-bentonite was investigated to determine the quality and purity of the samples, analysis conducted was settling rate, XRD, SEM and EDX analysis. After trying several purification methods, the as-received samples were analyzed before and after each trial, and the most promising method was the wet sieving, beside the methods that depend on dispersion into deionized

water. Therefore, an integrated method was designed for the process. The proposed method applies multi-stage sedimentation tanks and wet sieving to ensure maximum pure clay recovery. Before grinding, samples were put into oven for 16 hrs. at 105°C to get rid of the moisture. The weight before drying for one of the samples was 5 g and after drying, it became 4.6 g. This means that sample lost about 8% of its weight as water, which is an acceptable value, as the common water content in bentonite varies from 5% to 10%. Given the fact that, dispersion of solid particles is usually affected by particles size, and concentration. Therefore, at each sedimentation stage, and after sufficient retention time (about 5-10 minutes), most of the heavy and large sized non-clay particles will settle at the bottom of the first sedimentation tank. While leaving some other impurities still suspended, the supreme solution (about 90% of the height) is transferred to the second container, where water is added to decrease the concentration and enhance dispersion for further separation of impurities. At the first container, more water also will be added, and the process is repeated until a pure bentonite is obtained at the last stage, while at the first stage, only clear water at the top with solid non- soluble impurities at the bottom. Then the solution of pure bentonite will be wet sieved through 75 μm at recovery of more than 95%. Later, the wet sieved samples are dried to be grinded, and packed. A simple flow chart of the designed process is shown in **Figure 1**. The chemical composition was identified using X-ray diffraction (XRD), and Scanning Electron Microscopy (SEM).

2.4 Thermochemical upgrading and performance evaluation

The introduced thermochemical treatment in this paper is combination of Na_2CO_3 addition with heating (70 - 90°C) and stirring for a defined period. The heating causes bentonite platelet to expand and stirring give the water more chances to invade between bentonite platelets, then the particles swell when water be in-between. This swelling will remain after bentonite suspension is cooled down, because the water will be trapped between the particles. The thermochemical treatment method resulted in high water absorption and enhanced sodium activation process. The upgrading process was temperature and time optimized, and the rheological properties and filtration performance of the upgraded Ca-bentonite were evaluated before and after the treatment. Drilling fluid and rheological properties were tested according to API recommended practice 13-A. The sample was prepared using 22.5 gm of bentonite added to 350 ml of distilled water and agitated in high-speed mixer for 20 min. Then, the samples could age for 16 hours to ensure full hydration. At the beginning, the optimum dose of soda ash was defined, the treatment of raw Ca-bentonite was carried out at various doses of soda ash (Na_2CO_3), added to bentonite suspension of 6 wt.% as per API recommendation practice. The amounts of soda ash added was 2, 4 or 12 per 100 g of Ca-bentonite, and will be referred to as 2 wt.%, 4 wt.% and 12 wt.% Na_2CO_3 . Then the thermochemical upgrading technique was introduced to produce a high-quality bentonite for drilling fluid formulation. The heating temperature

was set at lower than 100 to avoid water evaporation, and the time was optimized by testing rheological properties and PSD of the samples after 1, 3, 6, 12, and 24 hours of heating and stirring.

The performance of upgraded Ca-bentonite was evaluated as a viscosifier and water control additive in real drilling formulas. Two drilling fluids formulas were selected based on field experiences and published data [24, 25, 26, 27, 28]. The two selected formulas, which are typically used in drilling operations was tested and compared with the same formulas that comprise commercial Na-bentonite.

3.0 RESULTS AND DISCUSSION

XRD analysis for raw and as-received Ca-bentonite is shown in **Figure 2**. The profile of intensity and 2θ degree showed the well-known peak of quartz at $2\theta = 26.5^\circ$, similarly other distinguished peaks was observed, such as montmorillonite, Illite, Calcite and Gibbsite. Generally, the XRD profile that generated for the local Ca-bentonite has the same pattern as typical bentonite pattern. SEM analysis conducted on bentonite plates showed layered orientation structure with small bumps at the surface and varying degree of roughness, whereas Ca-bentonite has surface structural features like the commercial Na-bentonite as shown in **Figure 3**. EDX provided a detailed chemical composition for Ca-bentonites and commercial Na-bentonite, the analysis is reported in **Table 2**. All clay distinguishing elements were present in comparable quantities such as those found in typical commercial bentonite, however, since it is originally calcium based, the content of calcium was higher, which resulted in low Na/Ca ratio. The analysis also showed existence of some other impurities such as Fe, Cu, Ti, and Zn, which is attributed to the erosion and weathering effects.

The composition of raw Ca-bentonite, is shown in the table 2. The purification method has reduced the levels of Fe, Cu and other impurities from as-received samples. The Ca-bentonites exhibited Na/Ca ratio of 0.915 to 1.97, which is considered a low ratio and is expected to affect water adsorption and swelling capacity. The commercial Na-bentonite exhibited a Na/Ca ratio of 2.89; therefore, it is more attainable to high swelling and good rheological properties, and qualified for drilling fluid applications. In addition, a considerable amount of magnesium was detected in Ca-bentonite samples, in the range of 0.7% - 0.9% by weight, which is higher than the amounts found in commercial Na-bentonite. However, magnesium is favorable in drilling fluids, since it can significantly improve gel strength and capability of cutting suspension in drilling fluids.

The beneficiation process has led to huge shift in particles size distribution (PSD). Analysis of as-received Ca-bentonite revealed a very wide range of particles sizes ($d_{10} = 1.92$, $d_{50} = 15.5$, $d_{90} = 1229 \mu\text{m}$), with average particle size (D_n) of $213.6 \mu\text{m}$. The value of D_n describes the number-average particle size

(ordinary arithmetic mean) calculated from equation (1), where D_i is the particle size with a frequency N_i .

$$D_n = \sum \left[\frac{N_i D_i}{N_i} \right] \quad (1)$$

However, after purification, and removal of impurities, the raw Ca-bentonite (purified) exhibited narrowed and uniformed PSD ($d_{10}= 0.96$, $d_{50}= 3.45$, $d_{90} = 10.42 \mu\text{m}$), with D_n of 4.9, which meets API specifications, and is almost matching with the PSD of the commercial Na-bentonite as shown in **Figure 4**.

3.1 Effect of chemical and thermochemical treatment

The chemical treatment with soda ash resulted in some improvement in the rheological properties. However, the major improvement was obtained after thermochemical treatment (upgraded Ca-bentonite). The chemical activation with soda ash has slightly enhanced the properties of raw Ca-bentonite and dispersion stability, however measuring of viscosity of 6 wt.% suspension indicated that improvement in viscosity was not sufficient. **Figure 5** Shows that viscosity increased with Na_2CO_3 addition up to 4 wt.% Na_2CO_3 in room temperature, while any further addition caused a reduction in viscosity values. The optimum soda ash amount was identified as the maximum amount needed to completely replace calcium in raw Ca-bentonite, and therefore is considered to be 4 wt.% of bentonite. Introducing of thermochemical treatment has significantly increased viscosity at low shear rate (1 s^{-1}) from 0.92 Pa.s to 4.1 Pa.s and at high shear rate (60 s^{-1}) from 0.033 Pa.s to 0.21 Pa.s, the effect of heating of chemically treated bentonite suspension at 70°C for 24 hours is shown in **Figure 6**.

3.2 Rheological measurements

Bentonite tends to form non-Newtonian fluids in water suspensions. The rheological properties of these suspensions depend on many factors, such as particles concentration and fluid temperature, in addition to the applied rate of shear. In drilling, water-based drilling fluids, which are formulated using bentonite, usually subjected to high stirring speed, then pumped through pipes of various diameters into the well. Inside the well, the drilling fluid will be subjected to different circulation speeds and pressures, beside the shear applied by rotation of the drilling string. Therefore, a flow sweep test was conducted on bentonite suspension samples before and after upgrading, to simulate these operational conditions. **Figure 7** shows viscosities at various shear rates ($0.1 - 100 \text{ s}^{-1}$), calculated from flow sweep test for raw, upgraded and commercial bentonite. Viscosity profile has shifted upwards with the treatment and showed higher viscosities at all shear rates.

The characteristic of bentonite suspension structure can be evaluated using dynamic frequency sweep test (Frequency oscillation sweep) at constant strain. Measurement was taken over a range of frequencies (100

to 0.1 rad/s) at a constant strain and at room temperature. The constant strain was selected within the linear viscoelastic region that is below the critical strain determined from a preceding strain sweep test. Analysis of raw Ca-bentonite and upgraded bentonite were compared with commercial bentonite results. **Figure 8** shows complex viscosities determined from the values storage modulus (G') and loss modulus (G''), which is related to drilling fluids capability of. The analysis showed very low values for raw Ca-bentonite before upgrading, which may cause hole cleaning problems. After treatment, the complex viscosity values of thermochemically upgraded Ca-bentonite have increased above commercial Na-bentonite, which is expected to enhance cutting lifting and hole cleaning.

Figure 9 shows the storage modulus (G') and loss modulus (G'') determined from the dynamic frequency sweep test, while strain was fixed at 10% to be within the LVR. The 6wt.% suspension of commercial Na-bentonite exhibited a storage modulus (G') higher than the loss modulus (G'') by a factor of 5. This suggests that the suspension structure is gel or solid-like, independent of the shear rate. Raw Ca-bentonite suspension showed high loss modulus over most of the frequency range, which will result in a liquid like behavior at all shear rates and it very low viscosity. For upgraded Ca-bentonite, the storage modulus has increased significantly from around 1 Pa to more than 150 Pa, and loss modulus increased from around 1 Pa to 50 Pa, while elasticity remained high with G'/G'' of ~ 3 . This great improvement reflects the impact of the thermochemical process for upgrading of Ca-bentonite, as shown in Figure 9 the improvement in rheology measurement is about 100 times the raw Ca-bentonite measurement. In drilling of oil and gas wells, the high storage modulus of the drilling fluid is an indication for hole cleaning. Higher storage modulus means good hole cleaning efficiency through better solid suspension and lifting of drilling cutting from the downhole to the surface, and preventing several drilling problems such as mechanical pipe sticking.

3.3 API drilling fluids testing

For drilling fluid testing, both in the laboratories and in the field, viscosities are measured using 6-speed dial viscometer. Experiments conducted on 8.6 ppg water-based drilling fluid, using raw Ca-bentonite showed very low readings at of 300 rpm and 600 rpm, these dial readings of viscometer are unite-less and viscosities are estimated by equations (2 to 4).

$$\text{Plastic viscosity, } PV \text{ (cP)} = \text{Dial reading at 600 rpm} - \text{Dial reading at 300 rpm} \quad (2)$$

$$\text{Apparent viscosity, } AV \text{ (cP)} = 0.5 \text{ (Dial reading at 600 rpm)} \quad (3)$$

$$\text{Yield point , } YP \text{ (lb/100ft)} = \text{Dial reading at 300 rpm} - \text{Plastic viscosity} \quad (4)$$

In addition, a very fast settling rates and poor dispersion was observed. The rheological properties depend on swelling of the clay due to water absorption, since raw Ca-bentonite has low swelling capacity due to the low Na/Ca ratio, its suspension did not show a significant increase in viscosity beyond water viscosity (1 cP). The dial readings of speed dial viscometer at 300 and 600 rpm were 2.8 and 5, respectively ($AV=2.5$ cP), which is very low when compared with the dial readings of 15 and 23 ($AV=11.5$ cP) for the commercial Na-bentonite.

However, after thermochemical treatment, values of speed dial viscometer for the upgraded Ca-bentonite has increased significantly from 2.8 to 35 at 300 rpm and from 5 to 40 at 600 rpm, which is much higher than the corresponding values (15 and 23) for commercial Na-bentonite. **Figure 10** shows the measured viscosities of the four samples: raw Ca-bentonite, chemically treated Ca-bentonite, upgraded Ca-bentonite and commercial Na-bentonite. The thermochemical treatment caused a major change in the structure of the Ca-bentonite suspension and as a result, the apparent viscosity increased from 2.5 cP to 20 cP, plastic viscosity increased from 2.2 to 6 and yield point from 1 to 28 lb/100 ft².

API filtration test was conducted for 8.6 ppg water-based bentonites mud at room temperature and 689.5 kPa (100 psi), and water loss was collected in a glass cylinder for 30 minutes. The water-bentonites system was prepared by adding 22.5 g of bentonite to 350 ml water and mixing for 20 minutes, then allowed to fully hydrate by aging for 24 hours. **Figure 11** shows that the raw Ca-bentonite exhibited very high water filtration loss; about 70 ml of water in 30 minutes. This water filtration amount is much higher than the API recommended value (~ 15 ml within 30 minute). The commercial Na-bentonite gave out 14.5 ml filtrate in 30 minutes. The large amount of water filtrates associated with raw Ca-bentonite is attributed to its very low swelling which resulted from weak water absorption and low Na/Ca ratio. The thermochemically-treated Ca-bentonite (treated with stirring and heating at 200°F i.e.93.3 °C) shows that the filtration gap between the untreated purified Ca-bentonite and commercial Na-bentonite has reduced significantly after activating the raw purified Ca-bentonite using soda ash, and being subjected to heating/stirring treatments for 24 hours. The thermochemically-treated bentonite yielded better filtration properties compared to the commercial one at the same conditions, after 30 minutes of filtration test; the cumulative water loss was 11.8 ml when upgraded Ca-bentonite was used.

3.4 Drilling fluid formulation using upgraded bentonite

Table 3 lists the weight percent and mixing times for the two selected drilling fluid formulas. The first Formula contains bentonite considered as a viscosity additive, caustic soda for pH control, xanthan gum as a viscosifier, and calcium carbonate as a bridging material [29, 30, 31, 32]. The density of this formula is

8.6 ppg (i.e. 64.33 lb/ft³) applicable for formation with normal pressure; other properties are shown in **Table 4**. The second formulation was prepared with barite, which is used to increase the weight of the mud. This formula is used to drill the top section of the well before reaching the oil/Gas reservoir section and must be altered before entering the oil formation (pay zone), because using of barite in oil zone will cause serious formation damage. The detailed formula is shown in **Table 3**. The formula contains 160 g of barite as a weighting material to increase the density of the drilling fluid. By adding barite to the drilling fluid, the density increased from 8.6 to 11.16 ppg (i.e. 83.48 lb/ft³), which can be used to drill formations with high pressures. Other additives are: caustic soda of pH adjustment and Carboxymethyl cellulose (CMC) for viscosity enhancement and to reduce the fluid loss of the mud. A Fann 35 speed dial viscometer was used for viscosity measuring of the mud after 24 hours aging to allow for complete hydration. The measurements are shown in **Figure 12**. The density of the mud was measured using mud balance (API, 1993), and other mud properties are listed in **Table 3**.

3.5 Thermal stability, hydraulics and hole cleaning

Thermal stability was performed by aging the drilling fluid for 16 hrs under two different temperatures 71°C (160°F) and 93.3°C (200°F). The results of viscometer dial readings are shown in **Figures 13** and **14**. The rheological properties were obtained using the Fan 35 viscometer at 48.9°C (120°F). **Table 5** lists the rheological measurements for commercial Na-bentonite and upgraded Ca-bentonite. After hot rolling at 71°C (160°F), the drilling fluid with upgraded Ca-bentonite has higher PV and YP compared to the drilling fluid prepared with commercial Na-bentonite. These results indicate the good thermal stability and hole cleaning capacity of the drilling fluid prepared with the upgraded Ca-bentonite (YP/PV = 1.02, which is within the range recommended by API-3A). After hot rolling at 93.3°C (200°F), the drilling fluid with upgraded Ca-bentonite has shown a higher yield point (1.15 times) in comparison with commercial Na-bentonite as presented in **Table 5**.

Moreover, at the YP/PV ratio of 1.24, better hole cleaning capacity is anticipated for the upgraded Ca-bentonite mud, especially for deep deviated and horizontal wells. The plastic viscosity of the drilling fluid with upgraded bentonite was very close to the one with the commercial bentonite, which means no additional requirement for pump specifications are required. Mahmoud [33] introduced an empirical correlation to illustrate the effect of yield point on the hole cleaning efficiency. Equation 5 can be used to determine the effect of changing the YP on hole cleaning efficiency.

$$\text{Hole Cleaning Efficiency} = 0.0047 (YP + 0.7815) \quad (5)$$

For example, assume 8.5" vertical hole diameter with a flow rate of 18.93 L/s (300 gpm), and the rate of penetration is 1.69×10^{-3} m/s (20 ft/hr), the drilling fluid with upgraded bentonite has a 100% cleaning efficiency using Eq. 5. Comparing the two drilling fluids, the upgraded Ca-bentonite will yield higher cleaning efficiency compared to the commercial one because it has higher yield point.

4.0 CONCLUSIONS

Based on the conducted analysis and the discussion presented in this study the following conclusions can be drawn:

1. An integrated and an efficient purification method and a novel chemical treatment for Ca-bentonite are developed in this study by heating and stirring the calcium bentonite in the presence of soda ash.
2. Ca-bentonite was effectively Na-activated using 4 wt.% soda ash. The treatment and upgrading method developed in this study altered Ca-bentonite properties and changed it from Ca-bentonite to Na-bentonite, which is the recommended type for drilling fluids additives. This level of improvement was not achieved in any of previous attempts to upgrade Ca-bentonite. Because of the thermal treatment, bentonite platelets expanded and resulted in high ion exchange process and water absorption, which significantly enhanced the rheological properties. The use of thermal treatment with stirring resulted in high quality bentonite compared to previous work.
3. The treated bentonite can be used in drilling fluids to drill the entire sections of the well and also can be used as a fluid loss control and viscofier in drill-in fluids. Also it can be used in oil and gas well cementing operations to reduce the cement slurry fluid loss.
4. The upgraded bentonite outperformed the commercial bentonite in rheological properties and exhibited higher viscosity, higher elasticity, lower filtration rate, and higher swelling capacity.
5. The upgraded Ca-bentonite was used in a typical water-based drilling fluid formula; the analysis showed that the rheological behavior is either similar or better of commercial drilling grade bentonite at low and high pressure/temperature conditions.

Nomenclature

- API : American petroleum institute.
 CMC : Carboxymethyl cellulose
 EDX : Energy-dispersive X-ray spectroscopy

LVR : Linear viscoelastic region
 OCMA: Oil company materials association
 SEM : Scanning electron microscopy
 XRD : X-ray diffraction

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Figure Captions List

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Table Captions List

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Table 1. API 13-A Drilling grade bentonite specifications

Property	Specified Limits
Viscometer dial Reading at 600 rpm	Minimum 30
Yield Point to Plastic Viscosity Ratio	Maximum 3
Filtration volume at 30 min	Maximum 15 cm ³
Residue after sieved with 200 mesh (75 μm)	Maximum 4.0% by weight > 75 μm
Moisture	Maximum 10.0% by weight

Table 2. EDX Analysis of Ca-bentonites and commercial Na-bentonite

Element	Commercial Na-bentonite	As-Received Ca-bentonite	Raw Ca-bentonite	Upgraded Ca-bentonite
O	58.65	54.52	54.61	55.04
Na	1.39	0.75	0.77	2.35
Mg	0.93	1.4	1.36	1.58
Al	6.83	9.16	10.49	9.87
Si	28.2	24.08	25.31	24.3
Cl		0.14	0.16	0.98
K	0.19	0.58	0.59	0.45
Ca	0.48	0.82	0.89	0.8
Ti	0.53	0.7	0.89	0.53
Fe	2.69	5.53	2.83	2.7
Cu		1.28	0.65	0.55
Zn		0.94	0.6	0.5
S	0.15			
Na/Ca ratio	2.89	0.915	1.97	2.95

Table 3. Drilling fluid formulations

Component, gm	Formula 1	Formula 2	Mixing Time, min
Water	350	250	-
Soda Ash	1	1	5
Bentonite	20	20	20
Caustic Soda	0.75	0.5	1
Xanthan Gum	0.75	-	20
High Visc CMC	-	1.5	10
Low Visc CMC	-	3.5	5
CaCO ₃	1	-	10
Barite	-	160	20

Table 4. Formula-1 Testing results for thermally treated Ca-bentonite and commercial Na-bentonite

Parameter	Commercial Na-bentonite	Thermally treated Ca-bentonite
Mud Weight, ppg	8.6	8.6
pH	12	12
Apparent Viscosity (AV)	32.5	30
Plastic Viscosity (PV)	20	10
Yield Point (YP)	25	40

Table 5. Testing results for Ca-bentonite and commercial Na-bentonite. Formula-2 @71°C and 93.3°C

Parameter	Commercial Bentonite		Local Upgraded Bentonite	
	71°C (160°F)	93.3°C (200°F)	71°C (160 °F)	93.3°C (200°F)
Mud Weight, ppg	11.16	11.16	11.16	11.16
Apparent Viscosity (AV)	63.5	63	68	68
Plastic Viscosity (PV)	41	41	42	42
Yield Point (YP)	41	45	43	52

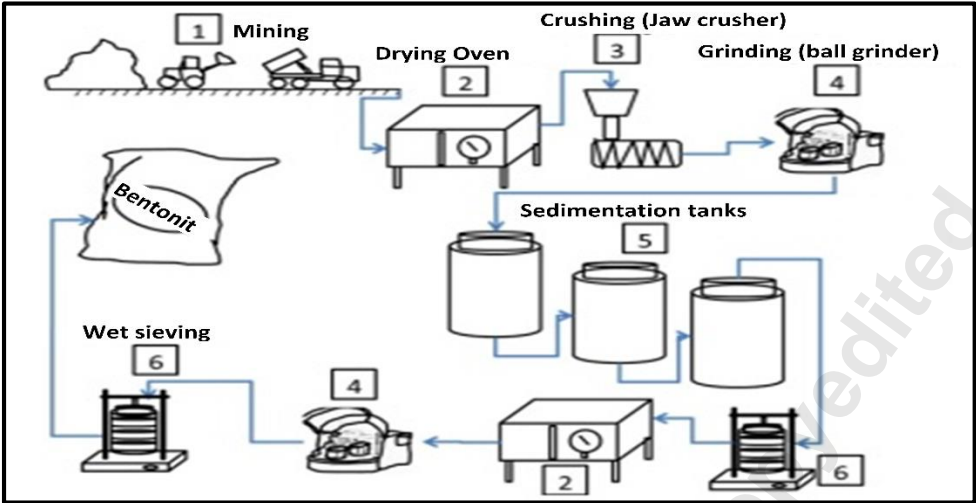


Figure 1. Purification process developed for as-received Ca-bentonite

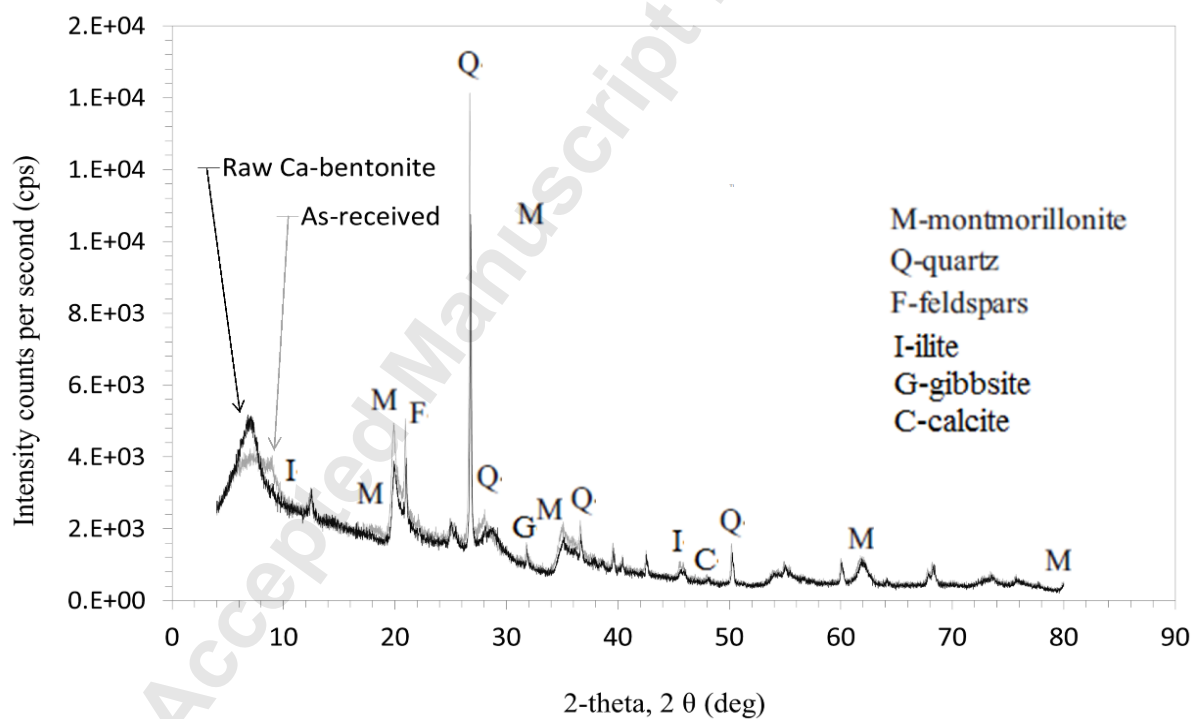


Figure 2. XRD pattern for as-received and raw Ca-bentonite (purified)

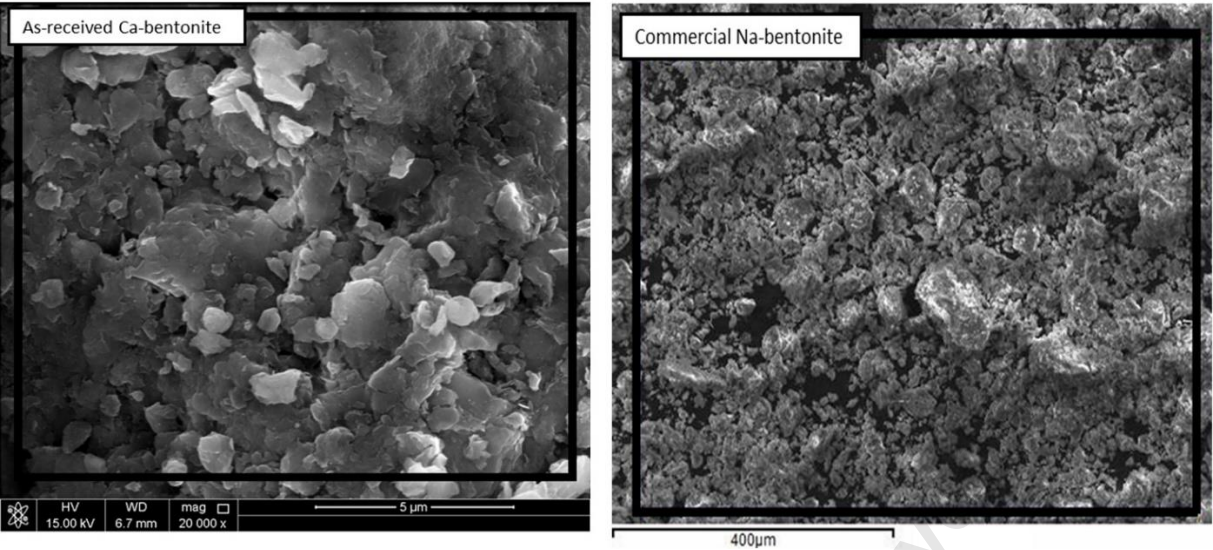


Figure 3. SEM images of as-received Ca-bentonite and commercial Na-bentonite

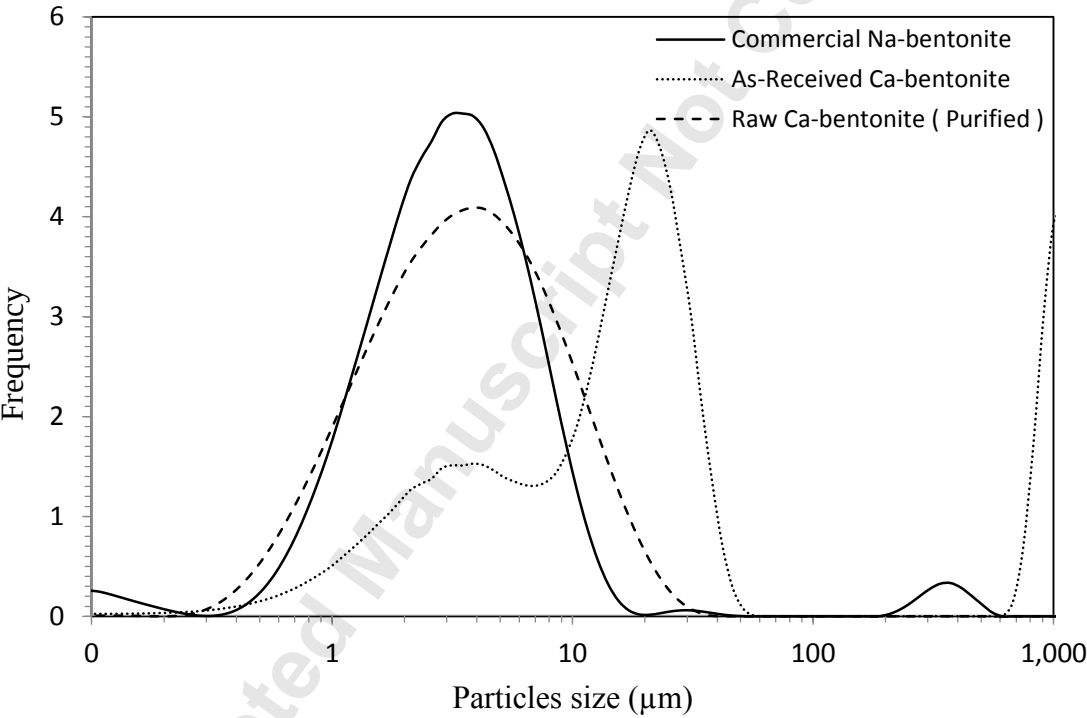


Figure 4. Particle size distribution for Ca-bentonites and commercial Na-bentonite

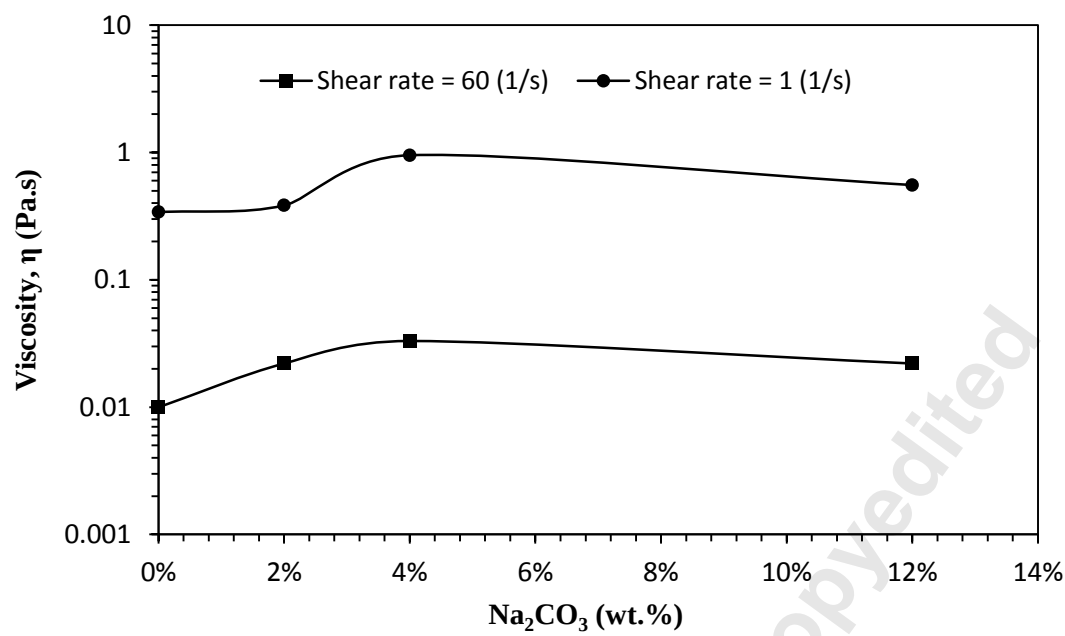


Figure 5. Effect of Na₂CO₃ addition on viscosity

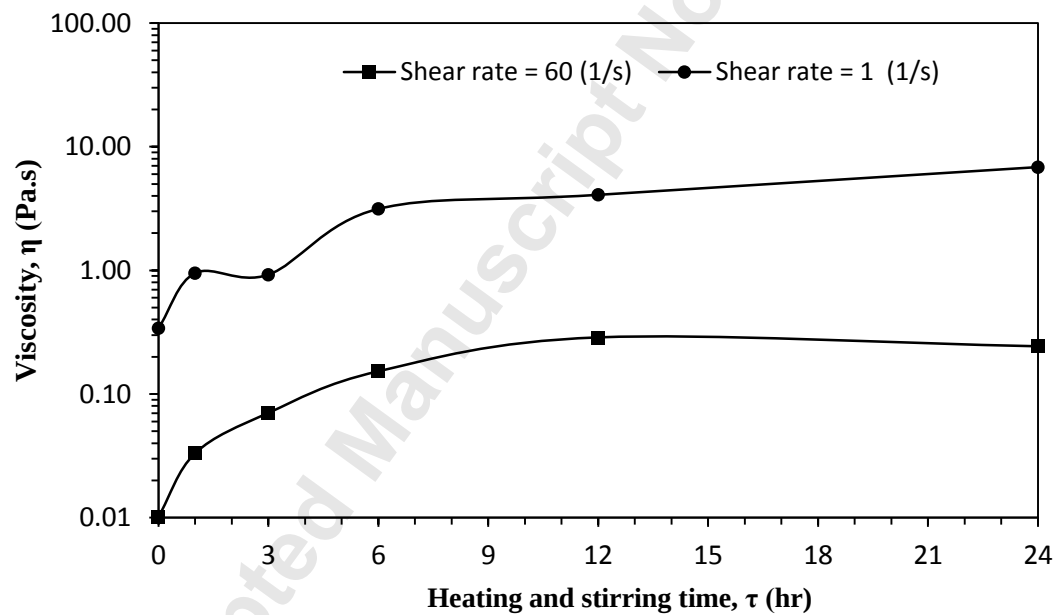


Figure 6. Effect of heating at 70°C on viscosity after 4wt.% Na₂CO₃ addition

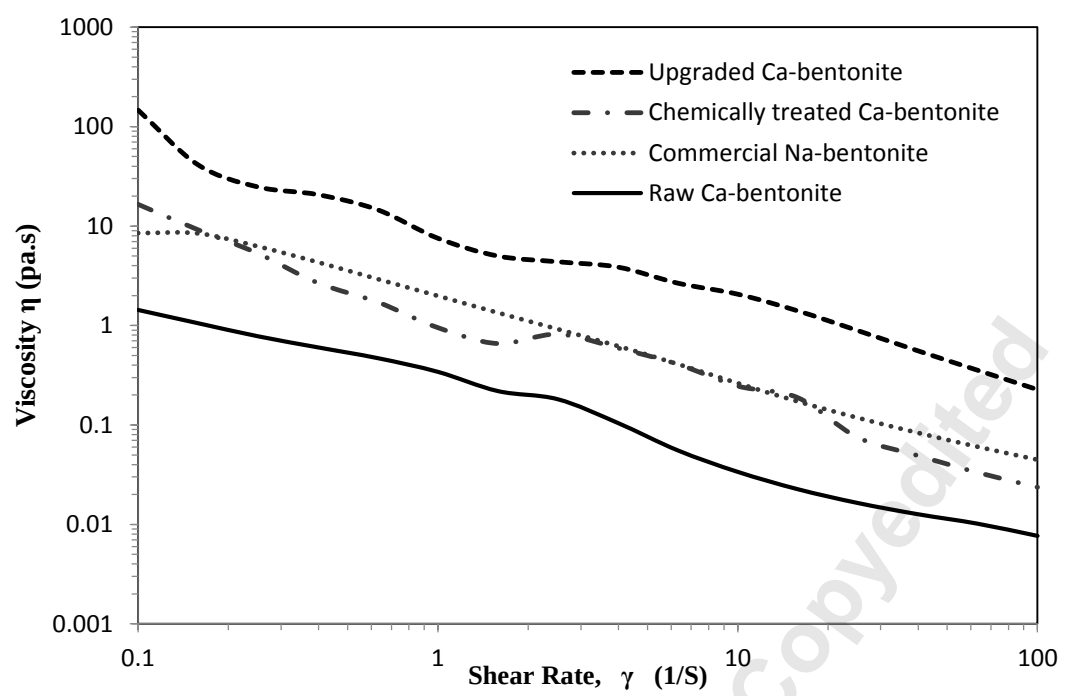


Figure 7. Viscosity over shear rate for Ca-bentonites and commercial Na-bentonite

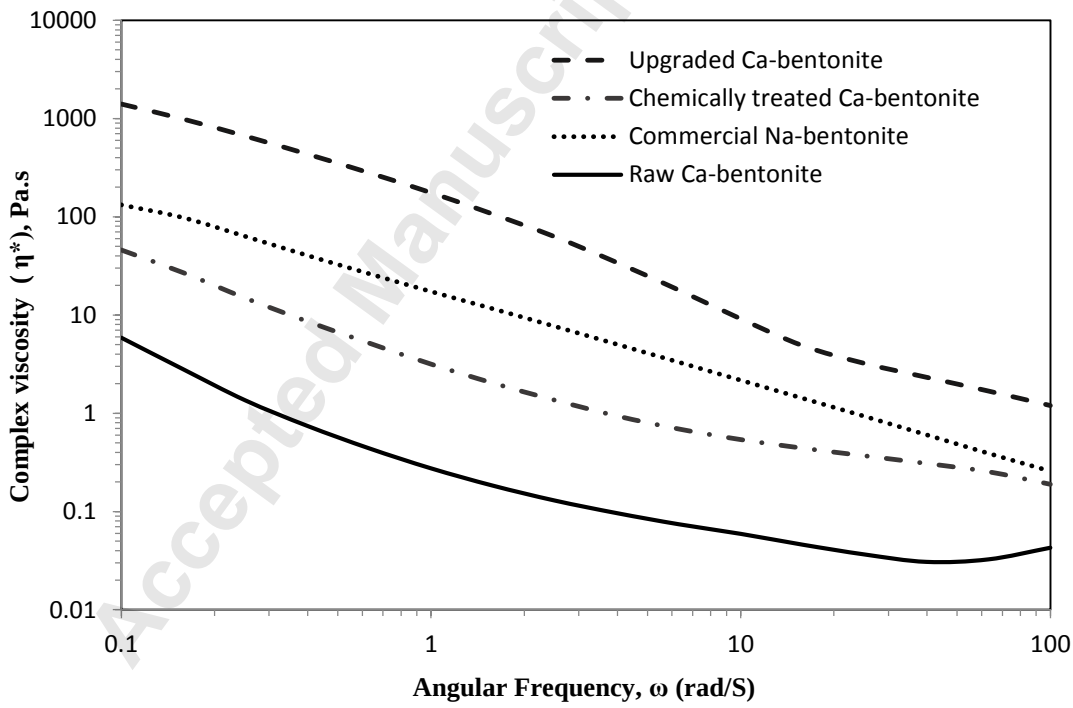


Figure 8. Complex viscosity for Ca-bentonites and commercial Na-bentonite

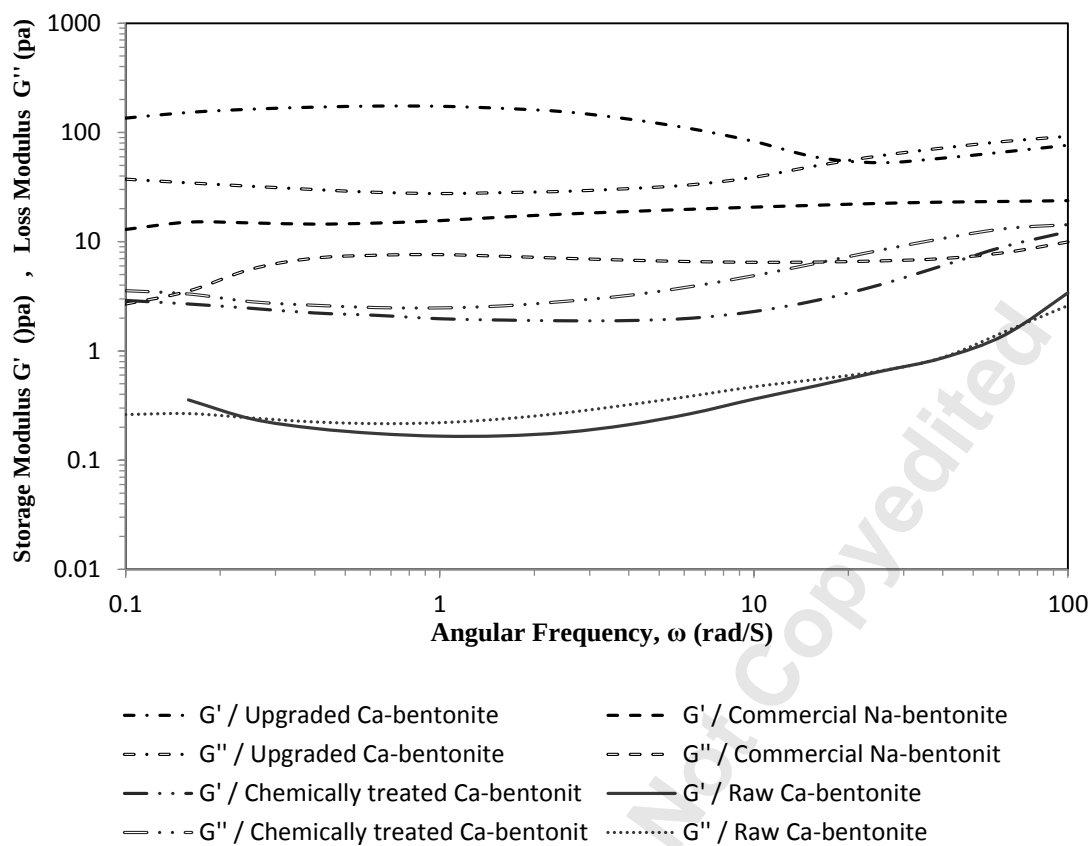


Figure 9. Elasticity by frequency sweep test for Ca-bentonites and commercial Na-bentonite

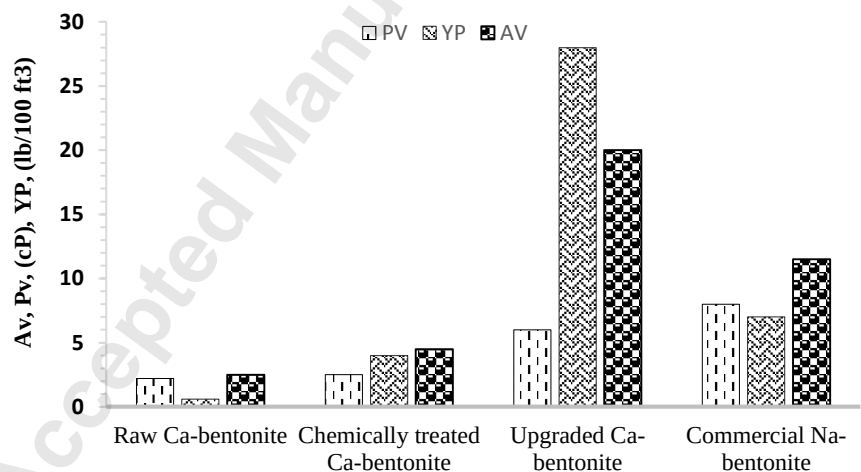


Figure 10. Viscosity and yield point for Ca-bentonites and commercial Na-bentonite.

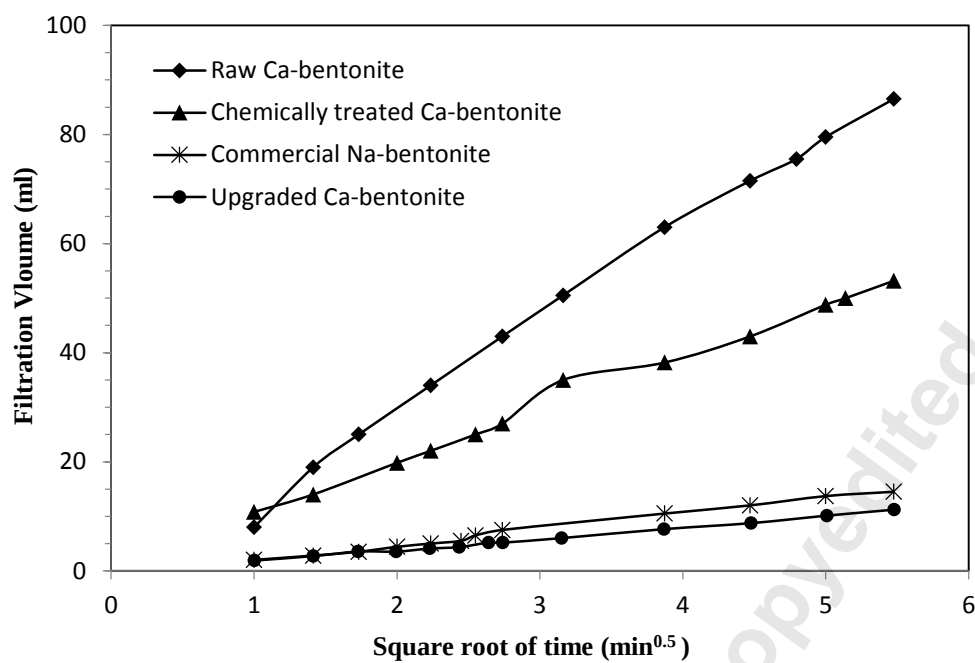


Figure 11. API filtration test at 100 psi and room temperature for bentonite mud

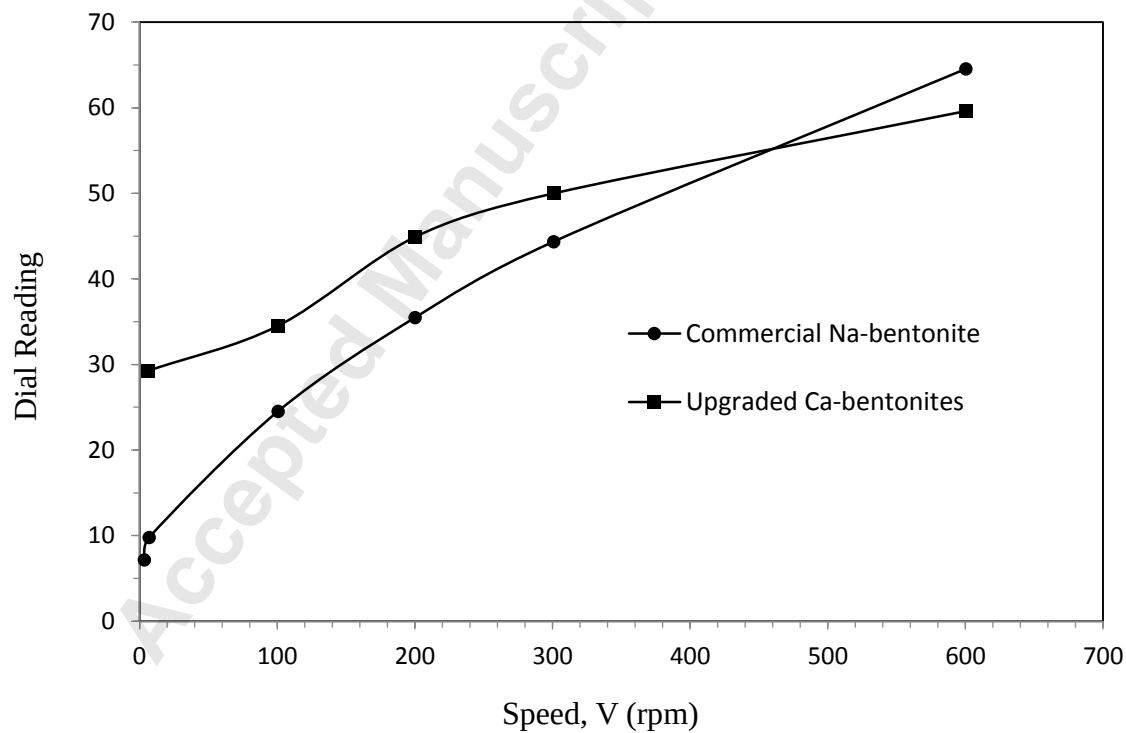


Figure 12. Formula-1 viscometer readings for thermally treated Ca-bentonite and commercial bentonite.

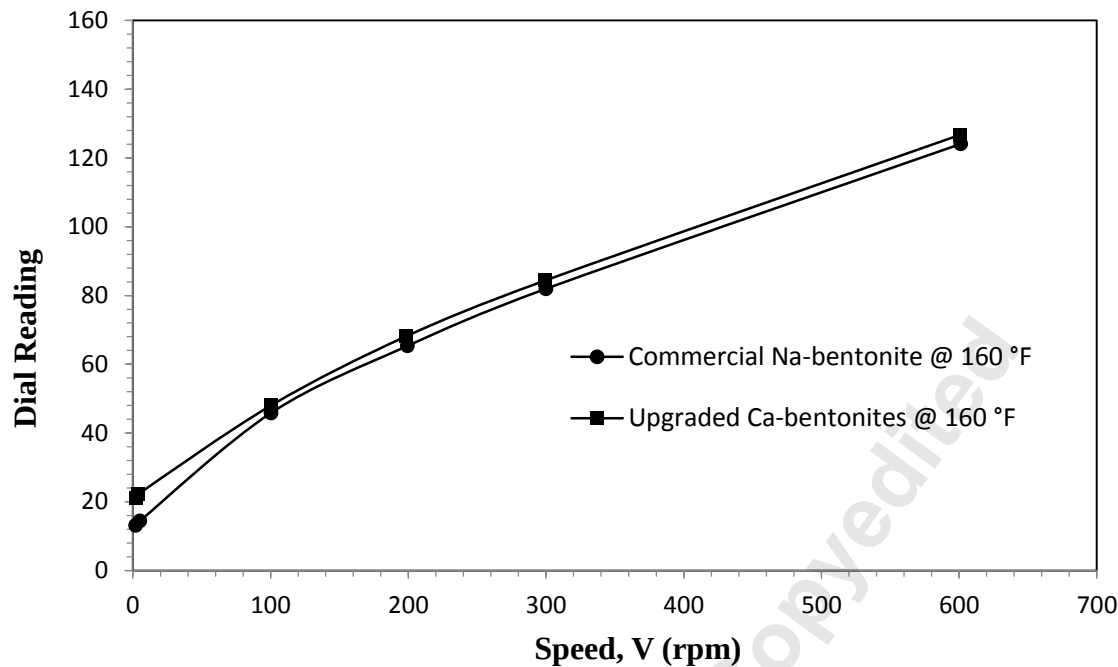


Figure 13. Formula-2 @ 71°C (160 °F) viscometer readings for thermally treated Ca-bentonite and commercial Na-bentonite

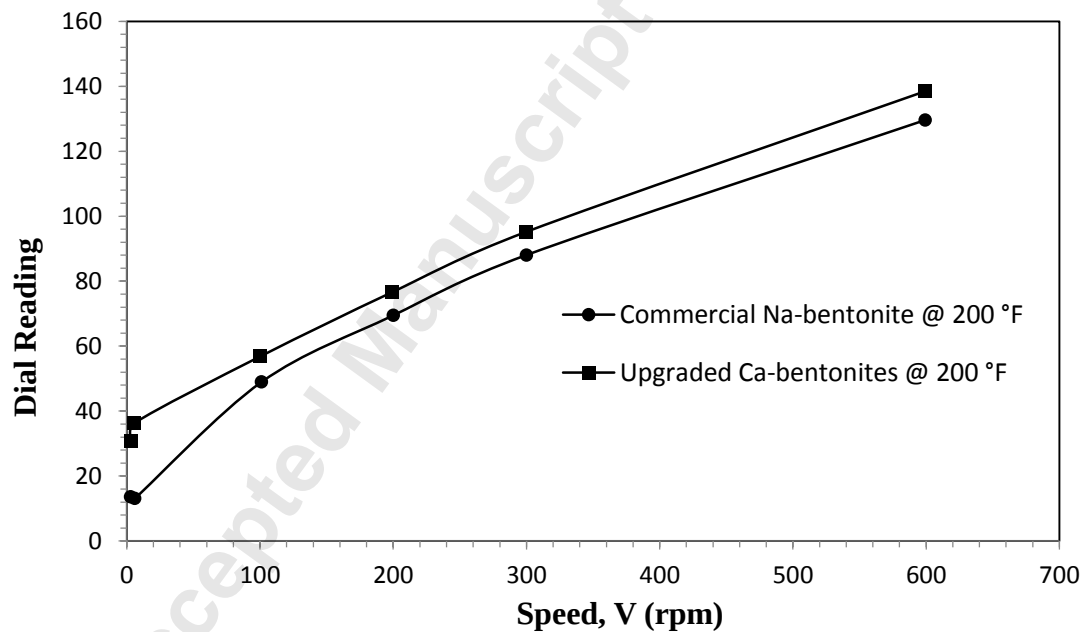


Figure 14. Formula-2 @ 93.3°C (200°F) viscometer readings for thermally treated Ca-bentonite and commercial Na-bentonite